MIL-I-82356/10
4 April 1969

MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES;
ALTIMETER PRESSURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356

1. SCOPE

1.1 This specification covers the detail requirements for simulated aircraft pressure altimeters for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-A-6863 | Altimeter, Pressure, 50,000 Feet, Type MB-1A |
| MIL-A-19679 | Altimeter, Pressure, Counter-Pointer |
| MIL-S-20708 | Synchros, General Specification For |
| MIL-A-23395 | Altimeter, Pressure, Counter-Pointer, Type MC-3 and MC-4 |
| MIL-A-27198 | Altimeter, Pressure AAU-7/A |
| MIL-A-27229 | Altimeter, Pressure AAU-8/A |
| MIL-A-27391 | Altimeter, Pressure AAU-2/A |
| MIL-A-38140 | Altimeter, Pressure AAU-16/A |

FSC 6930

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| | |
|---|---|
| MIL-I-82356 | Instruments, Simulated, For Aircraft Training Devices, General Specification For |

STANDARDS

Military

| | |
|---|---|
| MS 27572 | Case, Instrument, Flange-Mounted Aircraft |
| MS 27573 | Case, Instrument 2 3/4 Dial, With Sump Flange-Mounted, Aircraft |
| MS 25450 | Altimeter, Pressure, Counter-Pointer, Type MC-3 and MC-4 |
| MS 28044 | Altimeter, Pressure 50,000 Feet |
| MS 28074 | Altimeter, Pressure, Counter-Pointer |
| MS 33556 | Case, Aircraft Instrument, 3 1/4 Inch Size, Standard Dimensions For |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 The requirements of MIL-I-82356 and this specification shall apply.

3.2 Design basis.- Each simulated aircraft altimeter shall conform to the applicable operational instrument specified in Table I of this specification. In each case the activating component shall be coupled to the vernier pointer with a 1:1 ratio and the vernier shall drive the intermediate and main pointer with a ratio that will satisfy the applicable operational instrument display referenced in Table I.

3.3. Case.- With the exception of length, which shall not exceed the limits specified in the footnotes, cases shall be in accordance with Table I of this specification, as applicable.

3.4 Wiring.- Wiring shall be in accordance with the schematic referenced in Table I as applicable.

3.5 Barometric setting system.- A barometric setting knob identical in appearance and function to the barometric setting knob on the operational instrument shall be provided for each simulated instrument, as applicable.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 through 4.3.

4.2 Tests.- The following tests shall be performed on each instrument in addition to tests specified in the test requirements of MIL-I-82356:

(a) Ratio of 0 to 100 feet pointer to thousands and ten thousands pointers

(b) Backlash of gears

(c) Forward and reverse operation

(d) Running smoothness of gears

(e) No jumping or hesitation of pointers.

4.2.1 Electrical tests.- Electrical tests for instruments referenced in Table I shall consist of checks on electrical zero and running performance. Test equipment shall consist of a transmitter synchro of a size equal to or larger than that employed in the instrument actuator to be tested. The test transmitter synchro shall be rotated by a precision dial where zero degrees has been made to correspond to the reference line at the transmitter's true electrical zero in accordance with MIL-S-20708.

4.2.1.1 Electrical zero.- Tests for electrical zero shall be conducted with the transmitter synchro transmitter phase coordinated with the activated receiver synchro in accordance with MIL-S-20708. When the test transmitter synchro dial is at 0° ± 0.5 degrees, the activated synchro receiver shall read zero feet ± 5 feet.

4.2.1.2 Running tests.- Each receiver synchro, connected in proper phase compatibility as outlined in MIL-S-20708, shall be subjected to the following running tests, using the test set-up specified in 4.2.1:

(a) Slow run - The transmitter synchro shall be rotated clockwise and then counterclockwise at approximately 1 rpm for the full range of the indicator. The receiver synchro indicator pointers shall track smoothly without any discontinuities

(b) Fast run - The test of 4.2.1.2(a) shall be repeated in both directions at a rate of approximately one revolution per second. The display pointers shall follow without any discontinuities.

4.3 Activator ratio test.- The ratio between the pointers shall be such as to satisfy the requirements of design of the operational indicator.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.



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6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0027

TABLE I - SINGLE SYNCHRO DATA

| DESIG-NATION | OPERATIONAL INSTRUMENT TO BE SIMULATED | NOMINAL SIZE | *CASE TYPE | RATIO BETWEEN VERNIER, INTERMEDIATE AND MAIN POINTER | SCHEMATIC WIRING PER MIL-I-82356 FIG. NO. |
|---|---|---|---|---|---|
| -1 | MIL-A-19679**** | 3 inch | MS 28074 | 100 to 10 to 1 | 4 |
| -2 | MIL-A-23395**** | 3 inch | MS 25450 | 100 to 10 to 1 | 4** |
| -3 | MIL-A-27391 | 2 inch | MS 27572 | 100 to 10 to 1 | 4** |
| -4 | MIL-A-27229 | 3 inch | MS 27573 | 100 to 10 to 1 | 4 |
| -5 | MIL-A-38140 | 2 inch | MS 27572 | 100 to 10 to 1 | 4** |
| -6 | MIL-A-6863 | 3 inch | MS 28044 | 100 to 10 to 1 | 4 |
| -7 | MIL-A-27198 | 3 inch | MS 27573 | 100 to 10 to 1 | 4** |

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 6 inches without prior approval from the procuring activity.

** Designation denotes an integrally lighted instrument.

****These instruments to be activated with a size 15 synchro.



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